$\mathfrak{sl}_2$-action for q-analogues of multiple zeta values

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cf. Burmester's talk

$$\mathcal{G}^f$$

$$G^f(\mathbb{k})$$
formal MES

Thm. (Bachmann-Burmester ; '23)

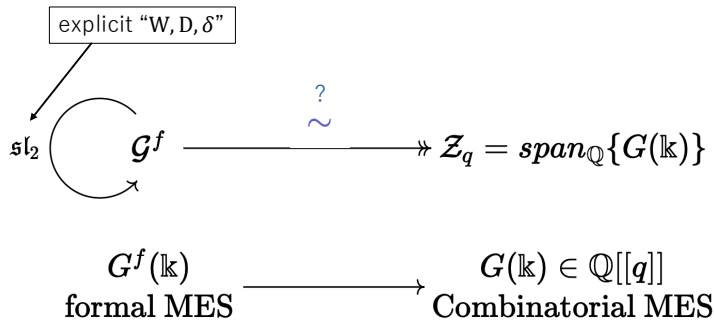
$$\mathcal{G}^f \twoheadrightarrow \mathcal{Z}_q = span_{\mathbb{Q}}\{G(\mathbb{k})\}$$

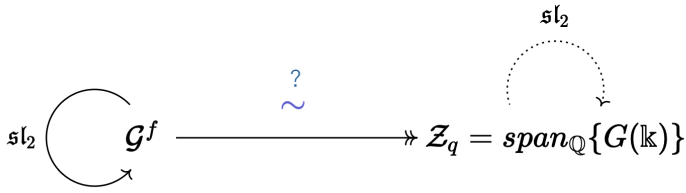
$$\begin{array}{c} G^f(\mathbb{k}) \\ \text{formal MES} \end{array} \longrightarrow \begin{array}{c} G(\mathbb{k}) \in \mathbb{Q}[[q]] \\ \text{Combinatorial MES} \end{array}$$

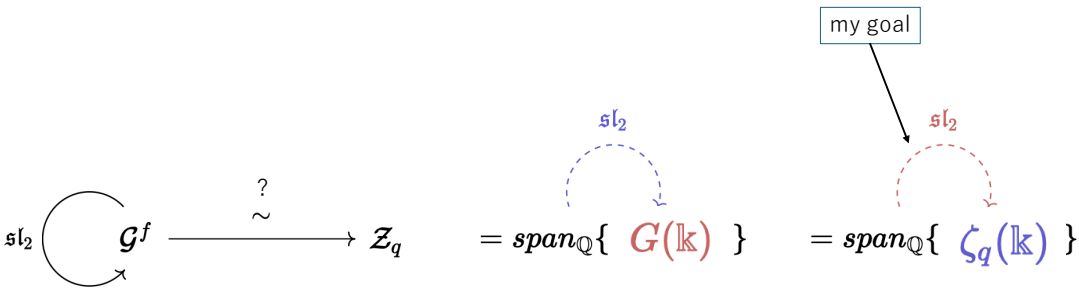
$$\mathcal{G}^f \overset{?}{\sim} \twoheadrightarrow \mathcal{Z}_q = span_{\mathbb{Q}}\{G(\Bbbk)\}$$

$$\begin{array}{c} G^f(\Bbbk) \\ \text{formal MES} \end{array} \longrightarrow \begin{array}{c} G(\Bbbk) \in \mathbb{Q}[[q]] \\ \text{Combinatorial MES} \end{array}$$

Thm. (Bachmann-Van Ittersum ; ‘24)




$\mathfrak{sl}_2$
$\mathcal{G}^f$
?
$\sim$
$\mathfrak{sl}_2$
$\mathcal{Z}_q = span_{\mathbb{Q}}\{G(\mathbb{k})\}$
$G^f(\mathbb{k})$
formal MES
$G(\mathbb{k}) \in \mathbb{Q}[[q]]$
Combinatorial MES
Problem : complicated and not explicit



blue : easy

red : hard

q-MZV

Definition (Schlesinger-Zudilin q-multiple zeta values)

For $\Bbbk = (k_1, \ldots, k_r)$ with $r \geq 1$ and $k_1 \geq 1, k_2, \ldots, k_r \geq 0$, define

$$\zeta_q(\Bbbk) = \zeta_q(k_1, \ldots, k_r) = \sum_{m_1 > \cdots > m_r > 0} \prod_{i=1}^{r} \frac{q^{m_i k_i}}{(1-q^{m_i})^{k_i}}$$

Denote $\mathcal{Z}_q := \mathbb{Q} + span_{\mathbb{Q}}\{\zeta_q(\Bbbk) \mid r \geq 1, k_1 \geq 1, k_2, \ldots, k_r \geq 0\}$.

The Eisenstein series are given by

$$G_k(q) = -\frac{B_k}{2k!} + \frac{1}{(k-1)!}\sum_{n \geq 1}\sigma_{k-1}(n)q^n = -\frac{B_k}{2k!} + \frac{1}{(k-1)!}\sum_{i=1}^{k}(i-1)!\left\{ {k \atop i} \right\}\zeta_q(i),$$

where $\left\{ {k \atop i} \right\}$ are the Stirling numbers of the second kind. In particular,

$$\mathbb{Q}[G_2, G_4, G_6] \subset \mathcal{Z}_q.$$

$\mathfrak{sl}_2$-triple

Definition

A triple (W, D, δ) of derivations is a $\mathfrak{sl}_2$-**triple** if

$$[W, D] = 2D, \quad [W, \delta] = -2\delta, \quad [\delta, D] = W$$

Example

The space of quasimodular forms $\mathbb{Q}[G_2, G_4, G_6]$ has an $\mathfrak{sl}_2$-triple.

$$D(G_k) = q\frac{d}{dq}G_k, \quad W(G_k) = kG_k, \quad \delta(G_k) = \begin{cases} -1/2 & \text{if } k = 2, \\ 0 & \text{otherwise.} \end{cases}$$

Question

We have $\mathbb{Q}[G_2, G_4, G_6] \subset \mathcal{Z}_q$. Can we extend this $\mathfrak{sl}_2$-action to $\mathcal{Z}_q$?

Algebra setup for q-MZV

Definition

- Let $\mathbb{Q}\langle a, b\rangle$ be the non-commutative polynomial ring in the variables a and b. We denote

$$\mathfrak{H}^0 := \mathbb{Q} + a\mathfrak{H}b \ \subset \ \mathfrak{H}^1 := \mathbb{Q} + \mathfrak{H}b \ \subset \ \mathfrak{H} = \mathbb{Q}\langle a, b\rangle$$

- $e_k := a^k b$ for all $k \geq 0$.
- And define the $\mathbb{Q}$-linear map ζ_q as

$$\begin{aligned} \zeta_q : \mathfrak{H}^0 &\longrightarrow \mathcal{Z}_q \\ e_{k_1} \cdots e_{k_r} &\longmapsto \zeta_q(k_1, \ldots, k_r). \end{aligned}$$

Algebra setup

Definition

Define the stuffle product $*_q$ for $w, u, v \in \mathfrak{H}^1$ and $k, l \geq 0$ as follows.

$$1 *_q w = w *_q 1 = w$$
$$e_k u *_q e_l v = e_k(u *_q e_l v) + e_l(e_k u *_q v) + e_{k+l}(u *_q v)$$

On $\mathfrak{H}^0$ define the involution τ by

$$\tau(e_{n_1} e_0^{m_1} \cdots e_{n_s} e_0^{m_s}) = e_{m_s+1} e_0^{n_s-1} \cdots e_{m_1+1} e_0^{n_1-1},$$

where $n_1, \ldots, n_s \geq 1$ and $m_1, \ldots, m_s \geq 0$.

Theorem

For any $w \in \mathfrak{H}^0$ we have $\zeta_q(\tau(w)) = \zeta_q(w)$.

Derivation $\widetilde{D}$

Theorem 3 (Singer; '15)

$\mathcal{Z}_q$ is closed under $D = q\frac{d}{dq}$. Moreover, we have

$$D\zeta_q(k_1,\cdots,k_r) = \sum_{i=1}^{r} k_i\Big\{(r-i+1)(\zeta_q(k_1,\cdots,k_r)+\zeta_q(k_1,\cdots,k_i+1,\cdots,k_r)) + \sum_{s=i}^{r}(\zeta_q(k_1,\cdots,k_s,0,k_{s+1},\cdots,k_r)+\zeta_q(k_1,\cdots,k_i+1,\cdots,k_s,0,k_{s+1},\cdots,k_r))\Big\}.$$

Definition (N.)

We define the $\mathbb{Q}$-linear map $\widetilde{D}$ on $\mathfrak{H}^0$ by

$$\widetilde{D}(e_{k_1}\cdots e_{k_r}) = \sum_{1\le i\le j\le r} k_i e_{k_1}\cdots(e_{k_i}+e_{k_i+1})\cdots e_{k_j}(e_0+1)e_{k_{j+1}}\cdots e_{k_r}.$$

Derivation $\widetilde{D}$

Theorem (N.)

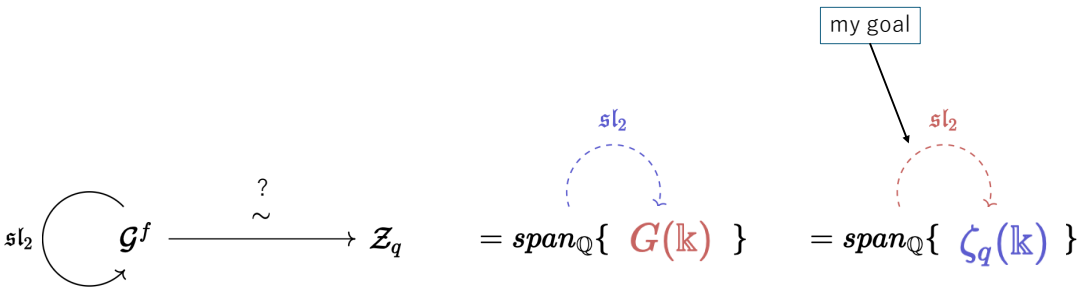
$\widetilde{D}$ is a τ-equivariant derivation for $(\mathfrak{H}^0, *_q)$ and satisfies $D \circ \zeta_q = \zeta_q \circ \widetilde{D}$.

The next step is to construct $\widetilde{W}$ and $\widetilde{\delta}$ so that $(\widetilde{W}, \widetilde{D}, \widetilde{\delta})$ is an $\mathfrak{sl}_2$-triple.

Depth 1 case

$$\widetilde{W}(e_k e_0^m) = (k+m)e_k e_0^m + k\sum_{i=1}^{k-1} \frac{(-1)^{i-1}}{i(i-1)} e_{k-i} e_0^m + (m+1)\sum_{j=1}^{m} \frac{(-1)^{j-1}}{j(j+1)} e_k e_0^{m-j} + \frac{(-1)^{k+m+1}(k(m+1)-1)}{2(k+1)(m+2)}$$

$$\widetilde{\delta}(e_k e_0^m) = \sum_{i=1}^{k-1}\sum_{j=1}^{m} \frac{(-1)^{i+j}}{i\,j} e_{k-i} e_0^{m-j} + \frac{(-1)^{k+m+1}}{2}(H_{k-1}H_m + \frac{H_{k-1}}{m+1} + \frac{H_m}{k})$$



blue : easy

red : hard